

Rutherford Scattering

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This study revisits the seminal Rutherford Scattering Experiment to further investigate the atomic structure using modern apparatus and data acquisition techniques. Through directed alpha particle bombardment on a gold foil, we aim to validate the inverse square law of scattering intensity relative to the angle and deduce the atomic number of gold. Employing a radioactive Americium-241 source, alpha particles with energies around 5.5 MeV are focused on a gold foil of $2.79\text{ }\mu\text{m}$ thickness. The scattering is measured at incremental angles from -5° to 20° , using a precision screw-adjusted detector within a vacuumed chamber to minimize interference.[1]

Our experimental setup incorporates updated digital data collection, utilizing LabView programs for automated signal acquisition and intensity histogram generation. Preliminary results demonstrate the expected decrease in scattered fractional intensity with increased scattering angle, in alignment with the Rutherford model. Furthermore, we determined a Z-value of 78.07 ± 4.14 , which falls well within the range of the expected Z-value of gold. A notable divergence from predicted values at larger angles hints at potential contributions from multiple scattering effects, nucleus-electron interactions, and other potential systematic uncertainties which goes to show the complexity of high-angle alpha particle scattering, but conclusively confirms the validity of the experiment.

I. INTRODUCTION

In the early 1900s, the atomic structure posed a puzzle until the initiation of scattering experiments involving alpha particles on metal foils by Geiger and Marsden at Cavendish Laboratory. A decade earlier, Thomson's identification of the electron and Millikan's subsequent determination of its charge had begun to unravel atomic enigmas. By 1909, the charge and mass of the electron were measured, dispelling uncertainties. Thomson's findings from X-ray diffraction on carbon indicated that an element's electron count correlated not with its atomic weight but with its atomic number, leading to the development of the periodic table. Due to the electron's minute mass compared to hydrogen, the lightest atom, it became evident that the majority of an atom's mass resides in its positively charged region.

The pursuit of understanding atomic structure further prompted the proposal that atoms are bound by resonating forces, emitting distinct spectral lines. This concept was partially supported by Balmer's earlier research, which mathematically explained hydrogen's spectral lines. Thomson, operating within the frameworks of Newtonian mechanics and Maxwell's electromagnetism, envisioned an atom as a sphere of positive charge with electrons scattered throughout, akin to raisins in a pudding. However, this 'raisin pudding' model, while intuitive, could not explain the precise spectral lines observed or the specific numerical patterns evident in hydrogen's spectrum and the more intricate spectra of heavier elements.

When Rutherford initiated his experiments on alpha particle scattering, the prevailing 'raisin pudding' model implied minimal deflections as the particles traversed the atom's assumed positive cloud. Nevertheless, the re-

sults of the experiment were unexpected: certain particles were deflected at large angles, a phenomenon that could not be accounted for by a uniformly charged sphere. Instead, these findings indicated the presence of a dense nucleus, which led Rutherford to propose a nuclear atom model. Helium nuclei, in the form of alpha particles, would only interact with the nuclei of gold atoms if they came into close proximity, a scenario that was statistically unlikely due to the nucleus's relatively small size. Despite this, the observed deflections suggested that when such interactions did occur, they resulted in significant scattering, ultimately giving rise to the development of the Rutherford scattering formula:

$$F(\theta) = F_0 e^{-\theta^2/\theta_m^2} \quad (1)$$

This equation describes how the fraction of particles deflected by an angle greater than θ decreases exponentially with increasing θ , where θ_m is the mean multiple scattering angle. Rutherford's formula was consistent with the experimental data for small deflection angles, but it diverged for larger angles due to the close proximity of the particles to the nucleus, resulting in a high probability of deflection.

According to Rutherford's findings, the intensity of the alpha particles scattered from a gold foil could be expressed as a function of the scattering angle θ and through experimentation, Rutherford's formula was validated (within uncertainty). This work laid the groundwork for modern scattering theory and fostered a new understanding of the atomic nucleus.

This laboratory exercise aims to replicate Rutherford's landmark experiment to reaffirm its conclusions. The primary objective is to demonstrate the inverse relationship between the intensity of scattered alpha particles and the

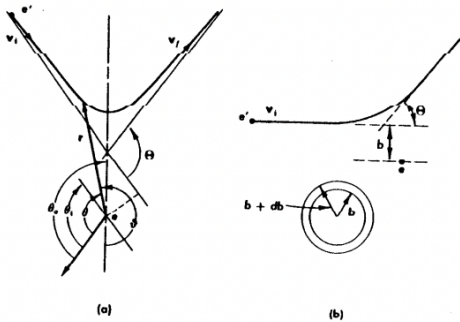


FIG. 1. Here we observe the encounter between two charged particles: a particle of charge e' approaching and then deviating from a stationary particle with charge e . The velocities before and after the encounter are denoted by v_i and v_f respectively, with the scattering angle labeled as θ . Subfigure (a) delineates the angles that play a role in the derivation of the scattering cross section. Subfigure (b) demonstrates the range of initial impact parameters b that lead to scattering into the differential solid angle $d\Omega$. (Figure 6.3b from Melissinos [2])

scattering angle, articulated as $\sin^{-4}(\frac{\theta}{2})$, and use this relationship to deduce the atomic number of gold. The experiment is significant not only as a historical reenactment but also as a practical application of scattering theory and angular analysis in physics. Our experiment will measure the distribution of scattering angles and the corresponding intensities to validate the Rutherford Scattering formula and, by extension, to calculate the atomic number Z of gold, thus mirroring the original methodology employed by Rutherford in his groundbreaking research.

II. BACKGROUND INFO AND THEORY

A. Deriving the Rutherford Scattering Formula

Alpha particles, which are well known for their ionizing effects, are composed of two protons and two neutrons, creating a particle with a double positive charge. Their diameter is similar to that of a helium nucleus, so around 1.67824 femtometers and thus in comparison to the entire atom, the gold nucleus is minuscule, as it is over 10,000 times smaller. This makes the probability of actually making direct collisions with alpha particles rare. During the interaction of an alpha particle beam with a group of nuclei, such as in a thin layer of gold, the particles scatter elastically, retaining their energy post-collision. This scattering is a result of the interaction between the charged spherical entities governed by the Coulomb force.

To derive the Rutherford scattering cross section $\frac{d\sigma}{d\Omega}$, we consider an alpha particle's trajectory under the influence of the Coulomb potential due to a nucleus. The key to the derivation is the conservation of angular mo-

mentum and energy in an elastic scattering process.

The force experienced by an alpha particle in the vicinity of a nucleus is given by $F = \frac{ZZ'e^2}{r^2}$, where Z and Z' are the atomic numbers of the alpha particle and the target nucleus, respectively, and e is the elementary charge. Using Newtonian mechanics, the equation of motion is

$$m \frac{d^2 r}{dt^2} = -\frac{dV(r)}{dr}, \quad (2)$$

with the potential energy $V(r) = \frac{ZZ'e^2}{r}$.

The conservation of angular momentum $l = mr^2 \frac{d\theta}{dt}$ and energy $E = \frac{1}{2}mv^2$ leads to the equation

$$\frac{d\sigma}{d\Omega} = \left(\frac{1}{4\pi\epsilon_0} \frac{Z_1 Z_2 e^2}{4E} \right)^2 \sin^{-4} \left(\frac{\theta}{2} \right) \quad (3)$$

which is the Rutherford scattering formula.

The derivation of the Rutherford formula involves integrating the equation of motion to find the relationship between the scattering angle θ and the impact parameter b . It incorporates the change of variables $u = \frac{1}{r}$ to solve the differential equation and obtain θ as a function of u , leading to the expression of the differential cross section in terms of the scattering angle. The full derivation of which can be found in Melissinos.[2]

In the scattering formula, Z_1 is the atomic number of the alpha particle, which is 2, and Z_2 is the atomic number of the material we shoot the beam into which is gold and 79. $\Omega = \frac{A}{r^2}$ is the solid angle for some spherical surface area A and radius r , and in spherical coordinates, $d\Omega = \sin\theta d\theta d\phi$. σ is the scattering cross-section angle, then the other parameters are ϵ_0 , the permittivity of free space, E , the kinetic energy of the alpha particle, and θ , the angle of the detector. [3]

B. Implications of Scattering

As θ approaches smaller values, the Rutherford scattering formula demonstrates divergence. This can be observed by the limit $\lim_{\theta \rightarrow 0} \sin^{-4} \frac{\theta}{2}$ approaching infinity. This mathematical behavior signifies an important phenomenon in physics: as the scattering angle decreases, the likelihood of scattering within the nucleus's cross-sectional domain increases. This amplifies the Coulombic interaction between the alpha particle and the nucleus. Graphically, this is represented by the impact parameter b approaching zero, indicating an indefinite intensification of the force between the two.

Furthermore, the trajectory of scattering is altered at angles beyond what the classical model predicts due to nuclear screening by the atomic electrons. This occurs because the nuclear charge is shielded by the electron cloud. The effect of screening becomes significant at angles greater than 30 degrees, causing a reduction in the effective charge encountered by the alpha particle. Consequently, this alters the expected scattering pattern.

In practical experiments involving gold foil, it is important to consider multiple scattering events. The probability of such events depends on the thickness of the foil relative to the alpha particles' mean free path. The mean free path refers to the average distance an alpha particle travels before interacting with another nucleus. When the foil thickness greatly exceeds the mean free path, the effects of multiple scattering become significant. In the case of a foil with a thickness of $2.79 \mu\text{m}$ and an alpha particle mean free path in gold of 38 nm , the probability of multiple scatterings is considerable.

When an alpha particle passes through a thin foil, it undergoes multiple Coulomb interactions with the atomic nuclei. These interactions can be approximated as small-angle scatterings. According to Molière theory, the root-mean-square scattering angle Θ_{rms} can be determined using the following formula:

$$\Theta_{\text{rms}} = \sqrt{\frac{4\pi N_A Z^2 e^4 t}{m_\alpha v^2 E X_0}} \cdot f(Z, \beta), \quad (4)$$

where:

- N_A is Avogadro's number.
- Z is the atomic number of the foil material.
- e is the elementary charge.
- t is the thickness of the foil.
- m_α is the mass of the alpha particle.
- v is the velocity of the alpha particle.
- E is the energy of the alpha particle.
- X_0 is the radiation length of the material.
- $f(Z, \beta)$ is a function that depends on the atomic number and the velocity of the particle relative to the speed of light $\beta = \frac{v}{c}$.

The function $f(Z, \beta)$ takes into account the screening effect due to the electrons surrounding the nucleus and is calculated from tabulated data for different materials.[4]

To calculate Θ_{rms} for a gold foil, we substitute the relevant values and expression into the equation. Doing so, yields:

$$\Theta_{\text{rms}} = \frac{13.6 \text{ MeV}}{v\beta z Z \sqrt{\frac{X}{X_0}}}, \quad (5)$$

This equation provides a good estimate for Θ_{rms} when the thickness of the foil is such that the particle undergoes many scatterings but not so many that the central-limit theorem would predict a Gaussian distribution for the scattering angles.

C. Atomic Number

The relationship between the scattering angle, the intensity of the scattered particles, and the differential cross-section is fundamental in understanding Rutherford scattering. The intensity of scattered alpha particles I_s is proportional to the original beam intensity I_0 , the area density of scatterers D , and the differential cross-section $\frac{d\sigma}{d\Omega}$:

$$I_s = I_0 D \frac{d\sigma}{d\Omega} d\Omega, \quad (6)$$

where D represents the area density of scatterers within the target material and is calculated as $D = \frac{t\rho D_0}{A}$. Here, t is the target thickness, ρ is the density of the target material, D_0 is Avogadro's number, and A is the atomic mass.

Upon isolating the differential cross-section, we obtain:

$$\frac{d\sigma}{d\Omega} = \frac{I_s}{I_0 D d\Omega}. \quad (7)$$

For Rutherford scattering, this differential cross-section is linked to the scattering angle θ and the atomic numbers of the alpha particle Z_1 and the target nucleus Z_2 as follows:

$$\frac{I_s}{I_0} = N d\Omega \left(\frac{Z_1 Z_2 e^2}{16\pi\epsilon_0 E} \right)^2 \sin^{-4} \left(\frac{\theta}{2} \right), \quad (8)$$

which simplifies to the following equation, representing the relationship between the scattering intensity and angle:

$$\frac{I_s}{I_0} = K Z_1^2 \sin^{-4} \left(\frac{\theta}{2} \right), \quad (9)$$

where K is a constant that incorporates all other constants and is determined to be $K = N d\Omega \left(\frac{e^2 Z_2}{16\pi\epsilon_0 E} \right)^2$. By measuring I_s and I_0 at different scattering angles θ , we can then determine the fractional intensity and fit the experimental data to this model and using this curve fit, we can extract the atomic number Z_1 of the alpha particle, thus verifying Rutherford's original findings.

It should be noted that despite the fact that alpha particles experience energy loss while traversing through a medium, we consider the average energy of alpha particles in our experiment to be 3.77 MeV . This value will be utilized in our calculations, which may introduce some degree of error. Furthermore, we disregard various sources of error in our experiment, including inelastic interactions, Cherenkov radiation, and bremsstrahlung.[3].

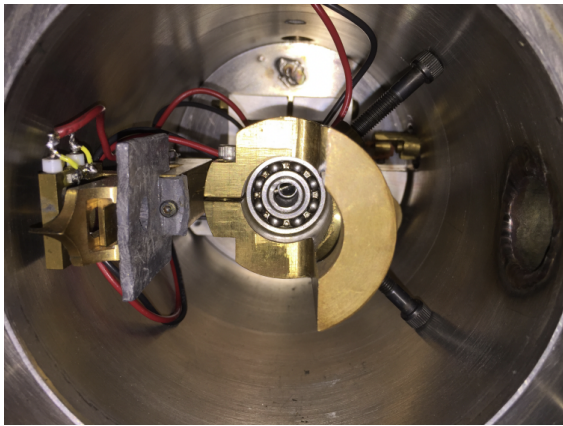


FIG. 2. This is a picture of the inside of the setup consisting of the plate where we would mount our gold foil, and a hole on the right where we shoot the alpha particles through, all including the alpha gun and the detector. [3]

III. EXPERIMENTAL OVERVIEW

A. Setup

In our experimental setup, we employ a radioactive source of ^{241}Am with an activity of $130\text{ }\mu\text{Ci}$, which emits alpha particles with energies around 5.5 MeV [3]. These energies correspond to the discrete energy levels from which the alpha particles transition, indicative of the quantum mechanical nature of the radioactive decay process.

The americium is deposited as a thin layer onto aluminum backing, ensuring that the emitted alpha particles have a well-defined initial energy as they escape the source. Subsequently, we employ a system of two collimating apertures designed to narrow the beam of alpha particles, thus providing a fine-tuned directionality toward the target, which in this case is a thin gold foil.

To capture and analyze the scattering data, a silicon detector is used, positioned at a variable angle relative to the incident alpha particles. The detector is interfaced with a computer running a LabView program, which records the number of alpha particle hits as a function of their detected energies. This data is represented in the form of a histogram, where each bin corresponds to a voltage range that is proportional to the energy of the alpha particle.

The LabView software also enables us to discern and quantify the scattering profile, as the number of counts at different energies reveals information about the angle of deflection the alpha particles experience upon interacting with the gold nuclei. By fitting the scattering distribution to theoretical models, we can extract significant physical parameters, including the cross-section of the interaction and the atomic number of the target nuclei.

Analysis of this data provides a modern confirmation of

the classic Rutherford scattering experiment, reaffirming the principles that led to the conceptualization of the nuclear model of the atom.

B. Procedure

Data acquisition is a critical component of our Rutherford scattering experiment. The setup involves securing the alpha source and the gold foil target in fixed positions within a vacuum chamber. To vary the detection angle, we manipulate the detector's position using a precision adjustment screw. The entire chamber is evacuated using a vacuum pump to minimize air scattering effects, ensuring that only alpha particles' interactions with the gold nuclei are recorded.

Prior to data collection, we ensure that the electronic systems, including the pre-amplifier and the Trans-L-Amp, are configured according to the guidelines provided in the literature [3]. The LabView's Soft Scope utility serves as an initial tool for observing the alpha particle signal, giving us a qualitative sense of the data we expect to collect. Once satisfied with the signal quality, we employ the PHA-5124 program within LabView to automate the data collection process.

As the scattering cross-section decreases with the increase in the scattering angle, it becomes imperative to collect a significantly larger dataset at higher angles. This compensates for the reduced count rates and ensures statistical reliability of our observations.

$$I(\theta) = I_0 \left(\frac{d\sigma}{d\Omega} \right) \Delta\Omega \quad (10)$$

$$F(\theta) = \frac{I(\theta)}{I_0} \quad (11)$$

Given that our apparatus might undergo physical adjustments over time, a calibration of the detector angle is essential. This involves establishing a zero-degree reference point to ensure accurate angle measurements during data collection. Additionally, we derive the fractional intensity $F(\theta)$ by normalizing the scattered intensity $I(\theta)$ to the flux of alpha particles emitted without the presence of a gold foil, thus necessitating a baseline measurement of this initial flux.

Upon completing the calibration and establishing the unscattered flux, we systematically gather data at 5-degree intervals, spanning from post-calibration zero up to 20 degrees, which is close to the detector's mechanical limits. To ensure the symmetry of the setup, we also record data at a negative angle of -5° , providing a sanity check for our experimental procedure.

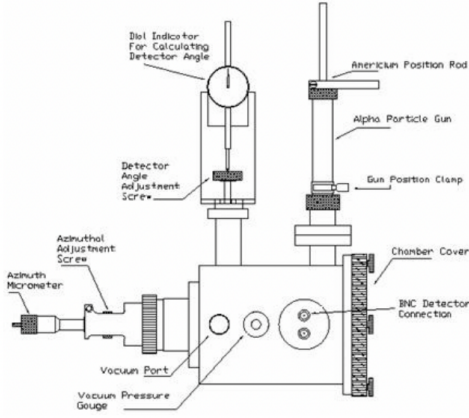


FIG. 3. This is a diagram of the full apparatus from the top view and from here we can see all the components of the setup. [3]

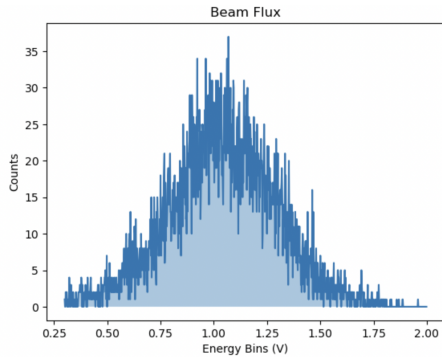


FIG. 4. This is a plot of the beam flux so the case where there is no gold foil, and we see a peak around 1.07 V.

IV. OVERVIEW PLOTS

We want to examine our data for the case where we have no gold foil first. To find the total intensity we multiply each bin count with the corresponding voltage then sum it all and divide by the total time we ran it to get Intensity (V/s). Here we get a total intensity of $185.039 \text{ V/s} \pm 13.307$.

Another important aspect of this lab is to examine the data, and generally looking at our data, we notice that for larger angles we get high amounts of noise, and thus we have extra peaks which creates errors in our data. After talking to the GSIs we determined that this is possible and it is smart to apply a cut and account for the removal of this data as error later on which we will do. So, when examining our data for 10, 15 and 20 degrees we apply a cut at around 1.15 volts as after that point for each graph there tends to be significant noise.

Meanwhile, our other plots have a singular peak before 1V and are skewed to the right as can be seen in our -5-degree and +5-degree runs.

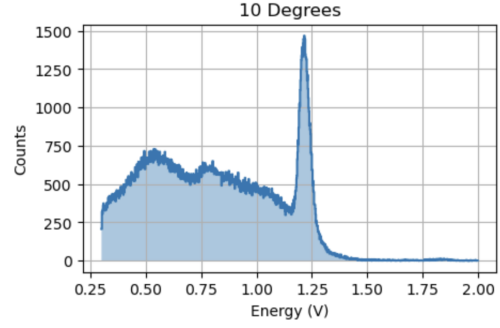


FIG. 5. Scatter plot of data from the detector at a 10-degree angle that we cut after 1.15V.

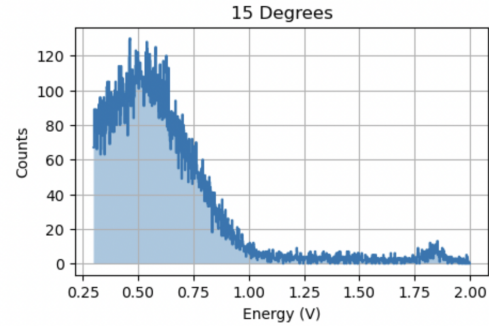


FIG. 6. Scatter plot of data from the detector at a 15-degree angle that we cut after 1.15V.

V. ANALYSIS

A. Calibration

Before taking analyzing any actual data, we need to calibrate our system, so we take small angle measurements around 0 degrees to determine the adjustment we need to make to properly calibrate to the true 0. By taking 1 degree increments around 0, we measured the angles from -3 to +4 degrees, and found the intensity of

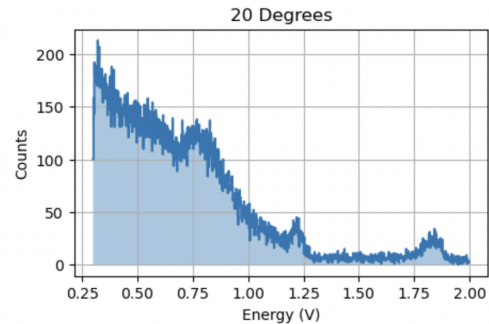


FIG. 7. Scatter plot of data from the detector at a 20-degree angle that we cut after 1.15V.

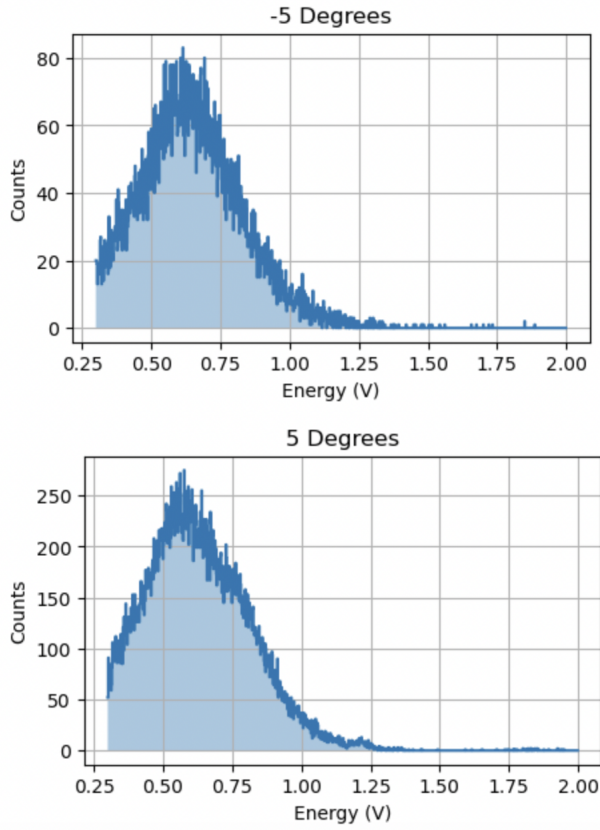


FIG. 8. Scatter plot of data from the detector at the positive and negative 5-degree angle for reference of a run that did not need to be cut.

each of these using the same method as before, where we multiply the bin counts of each bin by the voltage then sum this and divide by the total time that we ran it. Finally to get the fractional intensity we divide this intensity by the total intensity from the beam flux, and using this, we created a plot of the fractional intensities at each degree value.

Analyzing Figure 9, we clearly see a peak around 3 degrees, and we know that we should have a maximized intensity at the true 0 degree angle, so from this we can conclude that our detector is off by around 3 degrees, and by subtracting this from each of our measurements we will get the true degrees of measurement. So, after applying this calibration procedure to our measured angles, we create a plot of the true measured angles versus and their associated fractional intensities in Figure 10.

B. Results

From the previous section we decided to make cuts at 1.15V to refine our data for larger angles, but we will account for this error later. Thus, we decide to eliminate counts for voltages larger than 1.15V to refine our data as for angles larger than or equal to 10, our data is noisy and

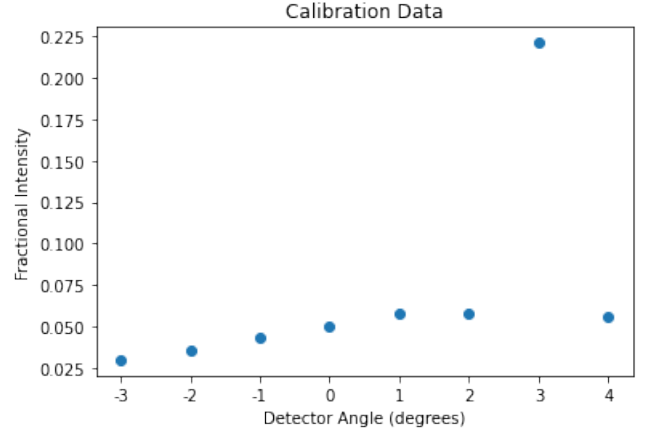


FIG. 9. Scatter plot that shows the fractional intensities of the detector at the angles from -3 to +4.

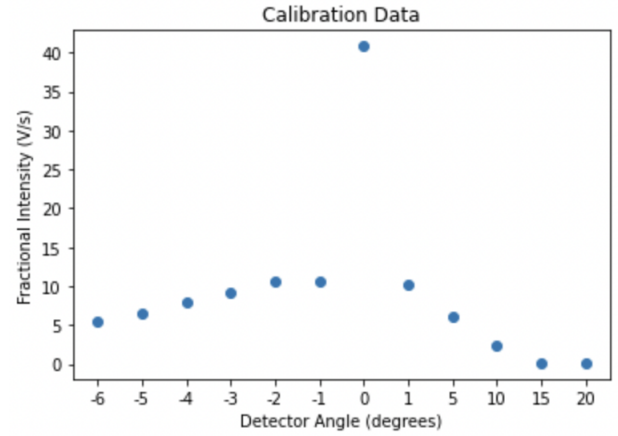


FIG. 10. Scatter plot that shows the fractional intensities of all the detector angle data after subtracting 3 degrees from all the measured angles.

spikes at high voltages. (We must remember to account for this in our errors as well)

After refining this data, we were able to use our prior methods of determining intensity to get the intensities for each of our degree points of -5, 5, 10, 15 and 20. Then using these intensities we divide by the total intensity we found earlier, which was 185.039 V/s, and this gives us all of our fractional intensities:

Angle (Degrees)	Fractional Intensity
-5	0.0347
5	0.0333
10	0.0123
15	0.00117
20	0.000738

TABLE I. Degrees and associated Fractional Intensities.

Now, using these fractional intensities as y coordinates and the degrees as x coordinates, we can plot these on a

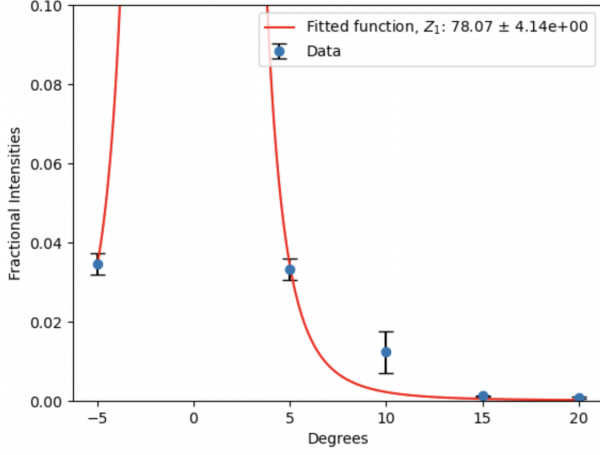


FIG. 11. The scatter plot of data from the detector at the prior mentioned degree values and the fit from Eq.(9) where we determine the Z value with error.

graph and model it with Eq.(9), and then using this fit, we can get our predicted Z for the experiment.

From this plot, we see that our Z value is within range of the atomic number of gold which is 79, so this makes sense, but we need to further our error analysis to determine the exact solutions.

C. Error Analysis

There are 2 uncertainties in the angle measurements, first due to the calibration since the true 0 is somewhere between 3 and 4 degrees, and an additional source from the granularity of the degree markings on the instrument. Each source of error contributes an uncertainty in the angle measurement, denoted as $\delta\theta_1$ and $\delta\theta_2$. These are both given as 0.5° .

To combine these uncertainties, we square each of them, sum the squares, and take the square root of the result. This is based on the Pythagorean theorem for the resultant of two independent error vectors (because errors can be considered as vectors in a multidimensional space of possible error directions):

$$\delta\theta_{\text{total}} = \sqrt{\delta\theta_1^2 + \delta\theta_2^2}$$

Since we're dealing with angles, we need to convert to radians for unitless factors so multiply by π radians for every 180° :

$$\delta\theta = \sqrt{\left(\frac{\pi}{180} \cdot \delta\theta_1\right)^2 + \left(\frac{\pi}{180} \cdot \delta\theta_2\right)^2}$$

Plugging in the values of $\delta\theta_1$ and $\delta\theta_2$ which are 0.5° each:

$$\delta\theta = \sqrt{\left(\frac{\pi}{180} \cdot 0.5\right)^2 + \left(\frac{\pi}{180} \cdot 0.5\right)^2}$$

Since both terms under the square root are the same, this simplifies to:

$$\delta\theta = \sqrt{\left(\frac{\pi}{180} \cdot 0.5\right)^2 \cdot 2}$$

Simplifying further:

$$\delta\theta = \sqrt{\frac{2}{180^2} \cdot \pi^2 \cdot 0.5}$$

This yields the uncertainty in the angle in radians when the angle is measured in degrees with a given error in the measurement, and we get a value of 0.0123.

The uncertainty in fractional intensity is determined by the distribution of arrivals, which is modeled by the Poisson distribution. For a Poisson-distributed variable N , the standard deviation is $\sigma = \sqrt{N}$. Let's denote the fractional intensity as $f = \frac{N_1}{N_2}$, where N_1 and N_2 are two independent Poisson-distributed variables representing counts, and their uncertainties are given by $\sigma_{N_1} = \sqrt{N_1}$ and $\sigma_{N_2} = \sqrt{N_2}$ respectively. The propagation of uncertainty for a function of two independent variables is given by

$$\sigma_f = \sqrt{\left(\frac{\partial f}{\partial N_1} \sigma_{N_1}\right)^2 + \left(\frac{\partial f}{\partial N_2} \sigma_{N_2}\right)^2}. \quad (12)$$

For a function defined as a ratio $f = \frac{N_1}{N_2}$, the partial derivatives are:

$$\frac{\partial f}{\partial N_1} = \frac{1}{N_2} \quad (13)$$

and

$$\frac{\partial f}{\partial N_2} = -\frac{N_1}{N_2^2}. \quad (14)$$

Substituting these into the uncertainty propagation formula yields:

$$\sigma_f = \sqrt{\left(\frac{1}{N_2} \sigma_{N_1}\right)^2 + \left(-\frac{N_1}{N_2^2} \sigma_{N_2}\right)^2}. \quad (15)$$

Given that $\sigma_{N_1} = \sqrt{N_1}$ and $\sigma_{N_2} = \sqrt{N_2}$, we substitute these into the equation:

$$\sigma_f = \sqrt{\left(\frac{\sqrt{N_1}}{N_2}\right)^2 + \left(-\frac{N_1}{N_2^2} \sqrt{N_2}\right)^2}. \quad (16)$$

This simplifies to:

$$\sigma_f = \sqrt{\frac{N_1}{N_2^2} + \frac{N_1^2}{N_2^4}}. \quad (17)$$

After factoring out $\frac{N_1}{N_2^2}$, we obtain:

$$\sigma_f = \frac{\sqrt{N_1}}{N_2} \sqrt{1 + \frac{N_1}{N_2}}. \quad (18)$$

Finally, by multiplying the numerator and the denominator inside the square root by N_2 , we have:

$$\sigma_f = \frac{\sqrt{N_1(N_1 + N_2)}}{N_2^3}. \quad (19)$$

This is the uncertainty in fractional intensity for a ratio of two Poisson-distributed variables N_1 and N_2 and to calculate this accurately we would also need the gain factor. [5] When particles pass through a detector, they may produce only a small amount of charge and the gain factor amplifies this signal so that it can be detected and measured by the electronics, and in this case our gain is of the factor 590, which we will have to divide out to map our raw data to actual data. Factoring this in and calculating the error values, we get:

Angle (Degrees)	Poisson Error
-5	2.360×10^{-5}
5	2.310×10^{-5}
10	1.393×10^{-5}
15	4.271×10^{-6}
20	3.386×10^{-6}

TABLE II. Angles and associated Poisson errors.

Now, let's write down the function for fractional intensity f based on the Rutherford scattering equation:

$$f(\theta) = K Z_1^2 \sin^{-4} \left(\frac{\theta}{2} \right)$$

To propagate the uncertainty from $\Delta\theta$ to Δf , we take the derivative of f with respect to θ :

$$\frac{df}{d\theta} = -4Z_1^2 \cdot \frac{K}{\sin^5(\frac{\theta}{2})} \cdot \left(-\frac{1}{2} \right) \cdot \cos \left(\frac{\theta}{2} \right)$$

Simplify this derivative:

$$\frac{df}{d\theta} = 2 \cdot K Z_1^2 \cdot \frac{\cos(\theta/2)}{\sin^5(\theta/2)}$$

The uncertainty in f (denoted as Δf) due to the uncertainty in θ (denoted as $\Delta\theta$) can be estimated as:

$$\Delta f \approx \left| \frac{df}{d\theta} \right| \Delta\theta$$

Substituting our expression for the derivative, we get:

$$\Delta f \approx \left| 2 \cdot K Z_1^2 \cdot \frac{\cos(\theta/2)}{\sin^5(\theta/2)} \Delta\theta \right|$$

This gives us our error in fractional intensity, and using this formula for each degree, we get:

Angle (Degrees)	Fractional Intensity Errors
-5	0.00271
5	0.00272
10	8.500×10^{-5}
15	1.123×10^{-5}
20	2.676×10^{-6}

TABLE III. Angles and associated Fractional Intensity Errors.

In our examination of the histograms depicting voltage distribution, we observed extraneous noise in the form of peaks at higher voltages, especially pronounced at larger angles of 10, 15, and 20 degrees. This phenomenon aligns with the expectations outlined by the GSIs and corroborates with the experiences reported by other groups. To mitigate the impact of these noisy peaks, we implemented a data cut-off at 1.15 volts. This introduces an additional error, termed 'cut error,' attributable to the anticipated extra intensity, which is typically represented by a constant background. By this method, we estimate the cut error based on the incremental intensity, which, in turn, affects the calculated fractional intensity, for which we get:

Angle (Degrees)	Cut Error
10	0.00519
15	0.000170
20	0.000189

TABLE IV. Cut errors for angles 10, 15, and 20 degrees.

As a final check to our error analysis, we can now calculate the chi-squared, by propagating these errors through, we get the final errors as:

Angle (Degrees)	Final Error
-5	0.00271
5	0.00272
10	0.00528
15	0.00182
20	0.00194

TABLE V. Final errors for angles -5, 5, 10, 15, and 20 degrees.

Now using these errors, we can find the chi-squared value, using the typical formula of observed minus expected divided by the associated error all squared for each

of the elements of the data, and furthering on that we can determine the reduced chi-squared, since we know there are 5 data points and 1 fit parameter, this gives us a degree of freedom of 4, so we multiply the chi-squared value by 0.25. From this, the chi-squared value (χ^2) is calculated to be 30.277, and the reduced chi-squared value (χ^2_ν) is 7.569. Generally even though the chi-squared value is really high indicating model discrepancy, it is not as important until we compare it to the number of degrees of freedom. Typically, a (χ^2_ν) close to 1 suggests a good fit between the model and the data. A value significantly greater than 1, like 7.569, could suggest that the model is not adequately describing the data, or that the uncertainties in the measurements might have been underestimated. In other words, the model's predictions deviate from the observed data more than what would be expected based on the reported errors, and thus there are likely other sources of errors.

In the Rutherford scattering experiment, several other sources of error were identified. The angle varies technically over the detector, and even though the variation is small, it can significantly affect the measurements at small angles. Unintended noise at large energies was also observed for some of the higher angles, although efforts were made to eliminate this noise. Another possible source of error is light entering the detector, which can interfere with the detection of scattered alpha particles. Moreover, there is an inherent energy loss of alpha particles as the energy used in the experiment is an average value; the alpha particles lose some of their energy when passing through the metallic capsule wall that encloses the source. Lastly, the total beam flux might not have

been entirely captured in the calibration step, adding to the list of potential errors in the experiment. All of this can account for the relatively high reduced chi-squared value which makes overall sense.

Finally, in the process of curve fitting, the `curve_fit` function from the `scipy.optimize` library was utilized to fit the model to the data, yielding both optimal parameter values and a covariance matrix. The square root of the diagonal elements of the covariance matrix gives the standard errors of the fitted parameters. Specifically, the error in Z_1 is obtained by taking the square root of the first diagonal element of the covariance matrix, representing the variance of Z_1 . Analyzing this error, we get a value of 4.14, so our final Z value is 78.07 plus or minus 4.14, and thus it is in range of the atomic number of gold and our conclusions and analysis make sense.

VI. CONCLUSION

In conclusion, we can see that we were able to successfully reproduce Rutherford's findings from his famous scattering experiment, as we were able to determine that the atomic number of gold is 78.07 with an error bound of 4.14, which is within range of 79, which is gold's true atomic number. Furthermore, from the error analysis we have seen that even though our fits may not be as accurate, there are many other systematic uncertainties that can be the cause for this, but in the end we can conclude that this experiment was successful.

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- [1] A. G. *et al.*, Two-photon frequency comb spectroscopy of atomic hydrogen, *Science* **370**, 1061 (2020).
 - [2] A. C. Melissinos and J. Napolitano, *Experiments in Modern Physics*, 2nd ed. (Academic Press, 2003).
 - [3] *Physics 111B: Advanced Experimentation Laboratory Manual – Rutherford Scattering*, University of California Berkeley, Berkeley, CA (2023).
 - [4] O. Voskresenskaya and A. Tarasov, *Moliere's multiple scattering theory revisited* (2014), arXiv:1204.3675 [hep-ph].
 - [5] E. Huttel, W. Arnold, H. Baumgart, and G. Clausnitzer, Screening corrections to the Rutherford cross section, *Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms* **12**, 193 (1985).

RUT - Rutherford Scattering

Signature Sheet

Student's Name Nishank Gite Partner's Name Mathi Raja

Before the Lab

You must watch the Radiation Safety video, and get a Radiation Ring before you use the apparatus in this experiment. View the Rutherford video about this experiment online.

Ring # 92842 Issued (Date and Signed) 02/15 [Signature] Returned (Date and Signed) _____

*Told to keep until
end of semester by
GSI in case we do
another radiation lab*

Pre-Lab Discussion Questions

It is your responsibility to discuss this lab with an instructor before your first day of your scheduled lab period. This signed sheet must be included as the first page of your report. Without it you will lose grade points. You should be prepared to discuss at least the following before you come to lab:

1. Describe Rutherford scattering. Read the following: Book by A. C. Melissinos, "Experiments in Modern Physics", Academic Press, New York, 2003, the section on Rutherford Scattering.
2. What is the Rutherford scattering formula? Define or explain each of the terms in the formula. What is a scattering cross section? How is the cross section related to what you actually measure in the experiment?
3. Why is it difficult to obtain data at large scattering angles?

Staff Signature [Signature] Date 02/16

Completed before the first day of lab? (Circle one) Yes / No

Mid-Lab Discussion Questions

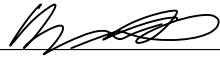
1. By day 4 you should have enough data to demonstrate the angular dependence of Rutherford scattering. Show it to an instructor and ask for a signature.

Staff Signature [Signature] Date 02/16

Completed by day 4 of lab? (Circle one) Yes No

Checkpoint Signatures

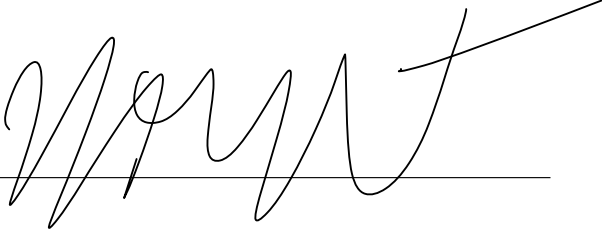
1. PN Detector

Staff Signature 

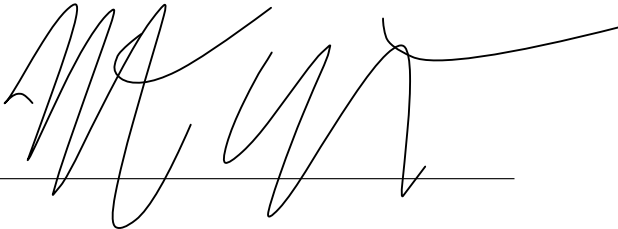
2. Signals

Staff Signature 

3. Measurements

Staff Signature 

4. Trial Run Analysis

Staff Signature 

5. Spectrum

Staff Signature 